

Exercise 9: Select fundamentals

SELECT AND FILTERING

SELECT AND SORTING

Question 1: Retrieve all columns from the employees table

SELECT *
FROM employees;

SELECT *
FROM workspace.default.employees;

id	first_name	last_name	department	Salary	hire-date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

Question 2: Find all unique departments

SELECT DISTINCT departments
FROM workspace.default.employees;

department
IT
HR
Finance
Marketing

Question 3: Retrieve first and last names ordered by salary descending

```
SELECT first_name, last_name  
FROM workspace.default.employees  
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Question 4: Retrieve the top 3 highest-paid employees

```
SELECT *  
FROM workspace.default.employees  
ORDER BY salary DESC  
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Question 5: Find employees in the IT department

SELECT

id,
first_name,
last_name,
department

FROM emp@workspace.default.employees
WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidu	IT

Question 6: Find employees in Finance with Salary > 60000

SELECT

id,
first_name,
last_name,
department,
salary

FROM workspace.default.employees
WHERE department = 'Finance'
AND salary > 60000

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

Question 7: Find employees in HR or Marketing

```
SELECT
    id,
    first_name,
    last_name,
    department
FROM workspace.default.employees
WHERE department = 'HR'
OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Question 8: Find employees not in IT

```
SELECT
    id,
    first_name,
    last_name,
    department
FROM workspace.default.employees
WHERE department <> 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Question 9: Find employees in IT, HR, or Finance using IN

SELECT

id,
first_name,
last_name,
department

FROM workspace.default.employees
WHERE department IN ('IT', 'HR', 'Finance');

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Question 10: Find employees in IT with salary > 65000
and city Johannesburg.

SELECT

id,
first_name,
last_name,
department,
Salary,
city

FROM workspace.default.employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg